

Folding a tessellated uniform-thick plate into compact stacks

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Abstract

Many large deployable rigid arrays for aerospace and civil applications, e.g., solar panels of satellites, reflectarray antennas, and foldable bridges, are preferably constructed from identical panels of uniform thickness. Achieving compact packing for easy transportation remains a significant challenge. This paper develops a mathematical framework to determine when a monohedral tessellated surface of uniform thickness can be rigidly folded into two compact stacks. We first divide the planar surface into a tessellation and define the cuts and folds along a Hamiltonian Circuit (HC) planned within the tessellation, producing a loop of panels. We then establish the necessary and sufficient conditions for the panels to be rigidly folded into two stacks. The first condition ensures that the panels can be folded into two equal stacks by making an imaginary cut along one edge and verifying whether the two ends of the resulting chain rejoin after an equal number of folds. The second condition requires the absence of entanglements or twists during folding when the loop is not cut open. The framework is validated through analytical examples and physical prototypes. This work advances the theoretical foundations of kirigami-inspired deployable structures and provides practical design guidelines for compact foldable planar structures.

Keywords: *kirigami-inspired deployable structure, Hamiltonian Circuit, rigid foldability, geometry and topology.*

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1 Introduction

Many large flat structures used in space and civil engineering applications must be folded into compact volumes for transportation and then deployed precisely into flat, continuous surfaces at their destinations. Examples include the solar panels of satellites [1-3], reflectarray antennas [4-6], and deployable bridges [7, 8]. Among the most widely used mechanisms for folding such rigid surface structures is the zigzag folding pattern, which expands and retracts unidirectionally [9]. Its simplicity and high packaging efficiency make it a preferred solution for many missions. When additional deployed surface area is needed, it is common to attach multiple stacks of folded panels to the main body of a spacecraft or satellite [10]. However, this approach leads to a deployed configuration resembling a long cantilever beam, which becomes increasingly flexible and mechanically vulnerable as the number of panels grows.

Alternative approaches have been developed to address these limitations. Recent origami-inspired thick-panel designs can form seamless top surfaces using careful geometric arrangements and supporting struts beneath the panels [11]. However, such systems typically require panels of different forms and non-uniform thickness, with the geometry of both the panels and supporting elements carefully designed and synchronised to achieve proper deployment. When using simple interlinked panels of uniform thickness, cuts are usually needed to accommodate the panel thickness during folding, transforming an origami model into a kirigami model [12-14].

Our previous study showed that an array of identical uniform-thick rectangular panels can fold into two compact stacks and deploy to a continuous and flat rectangular surface with no holes within the array (Fig. 1) [15, 16]. In such an array, each interior rectangular panel has four neighbours as illustrated by the shaded panel in Fig. 1(a), while panels along the edges of the array have fewer. When the array is folded into two compact stacks, each panel can only connect to two neighbours via revolute joints as one is folding onto its top face and one onto its bottom. Once those two joint connections are used, there is no space to fold the remaining neighbours, thus slits are introduced. The same applies to the rectangles on the edges of the array. Based on this observation, we trace a Hamiltonian circuit (HC) [17] in the array of panels. It is a path that begins and ends on the same panel such that each panel is visited exactly once. We place revolute joints where the circuit crosses two panels and cut slits elsewhere (Fig. 1b). A physical prototype was fabricated using this method, with its deployment process presented in Fig. 1(c). Subsequent efforts have applied the method to other geometries [18-20]. However, it is not clear which HC-based kirigami patterns can rigidly fold panels into equal stacks.

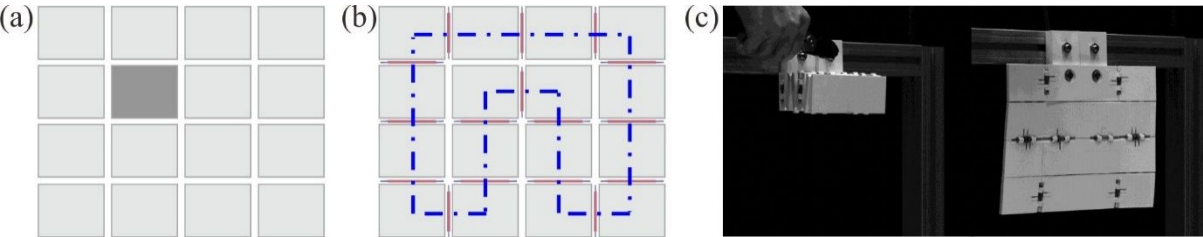


Fig. 1 Folding a rectangular array into a compact volume. (a) An interior panel in the array is initially connected with four neighbours. (b) A blue dotted dash line is plotted in the array to visit each panel exactly once. The connection remains if it is crossed by the path (thick red lines). A slit is made otherwise. (c) The deployment of a physical prototype [16].

In this work, we aim to fold any pre-defined flat surface with uniform panel thickness into a compact volume. Specifically, we develop a mathematical framework to tessellate the surface

using an even number of identical geometries such as squares, equilateral triangles, or regular hexagons that have reflection symmetry across their shared edges. This enables arbitrary curved boundaries to be approximated as stepped polygonal edges. Once tessellated, slits are introduced along selected edges, and the remaining edges serve as fold lines, allowing the entire array to be rigidly folded into two compact stacks. We establish the necessary and sufficient conditions for such foldability. Building on this framework, we also introduce an open-source programme to support practical engineering applications.

One might ask: why fold onto two stacks rather than one? Folding into a single stack requires all shapes to collapse onto a single panel, essentially forming a long, articulated open kinematic chain that behaves like a cantilever beam with many degrees-of-freedom (DoFs) and no inherent rigidity. Although such a configuration can still tile a plane when deployed, it is unsuitable for many engineering applications. Folding into two stacks represents the minimal configuration that reduces system's DoFs while offering practical mechanical stability. Designs that fold into any even number of stacks can be constructed by combining multiple copies of the two-stack patterns developed in this work. Folding into three or more stacks, while potentially interesting, is not addressed in this study and is left as a topic for future exploration.

The structure of this paper is as follows. Section 2 extends the HC approach to arrays of irregular outlines and other polygonal tiling. We partition any predefined surface into identical panels and then apply cuts and folds along their edges based on the circuit. Section 3 introduces the first condition to fold these polygons into two equal stacks: an imaginary cut along one edge transforms the closed kinematic chain into an open one, and if the cut rejoins after an equal number of folds from each end of the chain, the HC-based kirigami pattern is feasible. However, this condition alone does not prevent interference between non-adjacent panels during folding of the original closed kinematic chain. Therefore, Section 4 establishes the second condition: the folding must be untwisted and globally unobscured to ensure the rigid folding of the closed kinematic chain. Section 5 proves that these two conditions together provide the necessary and sufficient criteria for rigidly folding a loop of identical polygons into two equal stacks and illustrates their application through analytical examples and physical prototypes. Section 6 presents practical design strategies to reduce system DoF, thereby further enhancing the rigidity of the deployed surface. Finally, Section 7 concludes the paper and outlines directions for future research.

2 Extending HC approach to diverse geometries

In our earlier work, we applied the HC approach to fold an array of identical panels forming a rectangular outline (Fig. 2a) into two compact stacks. These panels, however, need not be limited to regular outlines; they can be arranged to form a wide variety of outlines in the deployed state (Fig. 2b–c). This prompts two questions: can the same approach be extended to fold irregular arrays into two equal stacks? More broadly, can we use the same approach if the array is composed of identical panels with other polygonal geometries?

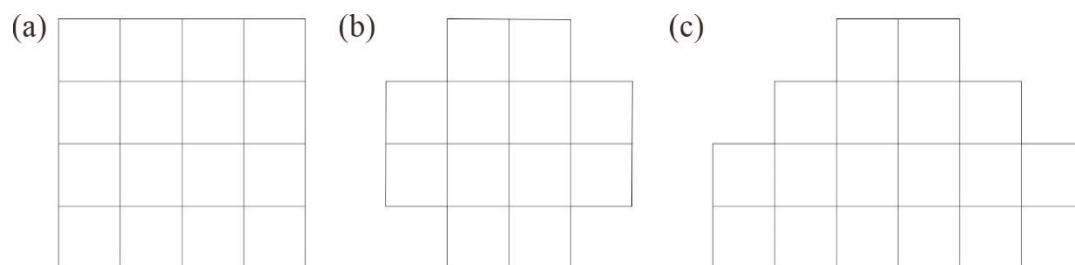


Fig. 2 Arrays with various outlines formed by identical rectangular panels.

2.1 Surface division, cuts, and folds

Now, we generalise the problem by considering transforming any pre-defined planar surface of uniform thickness into two compact stacks through geometric subdivision, making slits and folding. Specifically, this section addresses the following question: *Given a pre-defined planar surface of uniform thickness, how can we divide the surface and determine the locations of cuts and folds such that the entire assembly can be compactly folded into two equal stacks?*

2.1.1 Dividing a planar surface of uniform thickness through tiling

We begin by tessellating the planar surface into a periodic arrangement of identical polygons. Mathematically, this corresponds to a planar tiling [21], which partitions the plane without overlaps or gaps. Let $\mathcal{T}$ denote a tessellation consisting of congruent polygonal panels $\{p_i\}$.

Definition 2.1 (Regular Tessellation) A tessellation is regular if it consists of congruent regular polygons and each vertex has an identical arrangement of polygons. Only equilateral triangles, squares, and regular hexagons can produce regular tessellations (Fig. 3a-c) [22].

Beside regular tessellations, we consider derived forms such as half-equilateral triangles, half-squares, and parallelograms (Fig. 3d-f) that exhibit reflection symmetry along the shared edges.

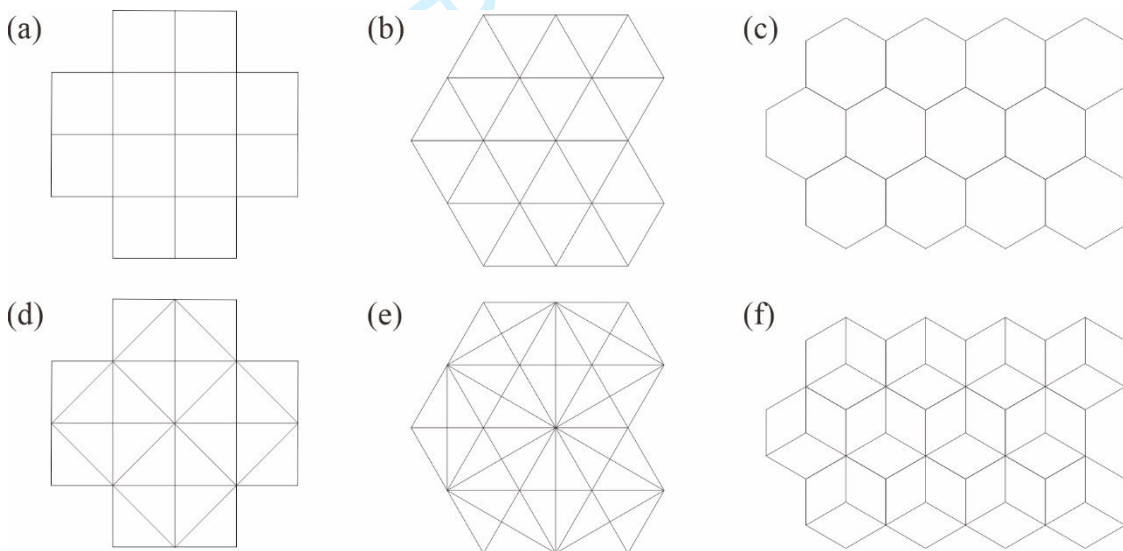


Fig. 3 Regular tessellations with (a) squares, (b) equilateral triangles, (c) regular hexagons, and monohedral tessellations with (d) half-square triangles, (e) half-equilateral triangles, and (f) parallelograms that exhibits reflection symmetry along the shared edges.

2.1.2 Graph representation and the kirigami pattern

A planar surface of uniform thickness must be divided into an even number of polygonal panels with reflection symmetry across their shared edges to enable folding into two compact stacks. Reflection symmetry ensures that adjacent facets can fold along their common edge and align precisely, allowing for complete overlap of facets. Moreover, having an even number of panels guarantees that each panel can be paired between the top and bottom layers of the stacks.

Definition 2.2 (Grid Graph) A grid graph $\mathcal{G} = (V, E)$ is an undirected graph [23] formed from a tessellation $\mathcal{T}$ by assigning a node $v_i \in V$ to each polygonal panel p_i and an edge $e_{ij} \in E$ between nodes v_i and v_j if the corresponding panels in the tessellation share a common edge. Example grid graphs for regular tessellations are shown in Fig. 4(a-c).

A tessellated surface of uniform thickness can be regarded as an assembly of rigid panels that are connected via revolute joints. It is analogous to an origami model where each rigid panel is a facet whereas revolute joints are creases. The revolute joints connect the panels on the top or bottom facet depending on if it is a valley or mountain crease, respectively, in an origami model.

The number of neighbouring panels depends on the geometry of tiling. For example, in a regular tessellation, each square is connected to four neighbours, an equilateral triangle to three, and a hexagon to six. While such connectivity gives a continuous, hole-free surface in the deployed state, it becomes problematic when the structure is folded into two compact stacks. In this folded configuration, each panel must be connected to exactly two others, with one folding over it and one folding underneath. This implies that a subset of the original connections must be removed to avoid over-constraining the assembly, turning an origami model to a kirigami model, where some creases are replaced by slits. To guide this process, we extend the HC approach introduced in Section 1 to more general tiling patterns. By tracing a HC through the grid graph (Fig. 4d-f), we can obtain a valid folding configuration: the circuit visits each panel exactly once, ensuring that each panel connects to exactly two neighbours. The edges crossed by the HC are creases, while the others are slits (Fig. 4g-i).

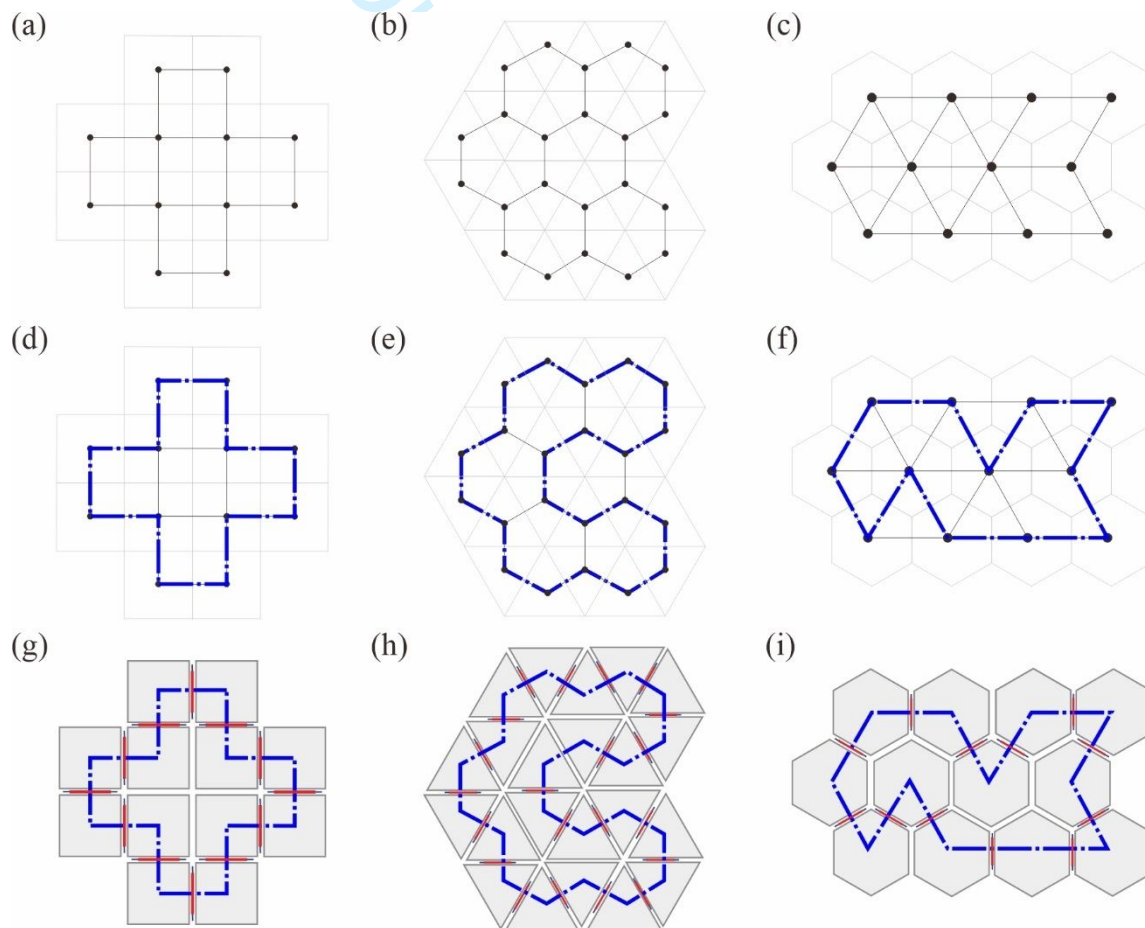


Fig. 4 Grid graphs in the arrays of (a) squares, (b) equilateral triangles, and (c) regular hexagons. (d)-(f) A HC is plotted in blue dotted dash lines within each grid graph. (g)-(i) Folds or revolute joints are placed according to the HC and denoted by thick red lines.

Proposition 2.1 A grid graph $G = (V, E)$ can be derived from a tessellated surface. A Hamiltonian circuit $H \subseteq E$ identifies a valid assignment of creases and slits such that each panel connects to exactly two neighbours, thereby providing feasibility of folding into two compact stacks from a continuous surface.

Remark 2.1 The existence of such a Hamiltonian circuit depends on the underlying geometry and parity of the number of panels. In rectangular and hexagonal tessellations, HCs are often guaranteed when the number of nodes is even [25, 26], whereas in triangular arrays, the existence of HCs is not guaranteed [27]. Finding a HC is NP-complete [28], and brute-force enumeration remains the most practical method [29]. We use Python code by Sharma [30] to enumerate all HCs in G and then select all non-isomorphic ones [31].

Example 2.1 There are usually many non-isomorphic HCs within a grid graph. For example, many HCs can be found through exhaustive search in a 6×5 rectangular array; six of them are shown by the blue dotted dash lines in Fig. 5. If an edge is crossed by the path of an HC, a revolute joint is placed on either the top or bottom surface of the panel; if not, a slit is placed, as highlighted in the figure. Among the many possible ways to arrange the slits, it is not obvious if the slit arrangements can result in a two-stack foldable configuration, which is discussed next.

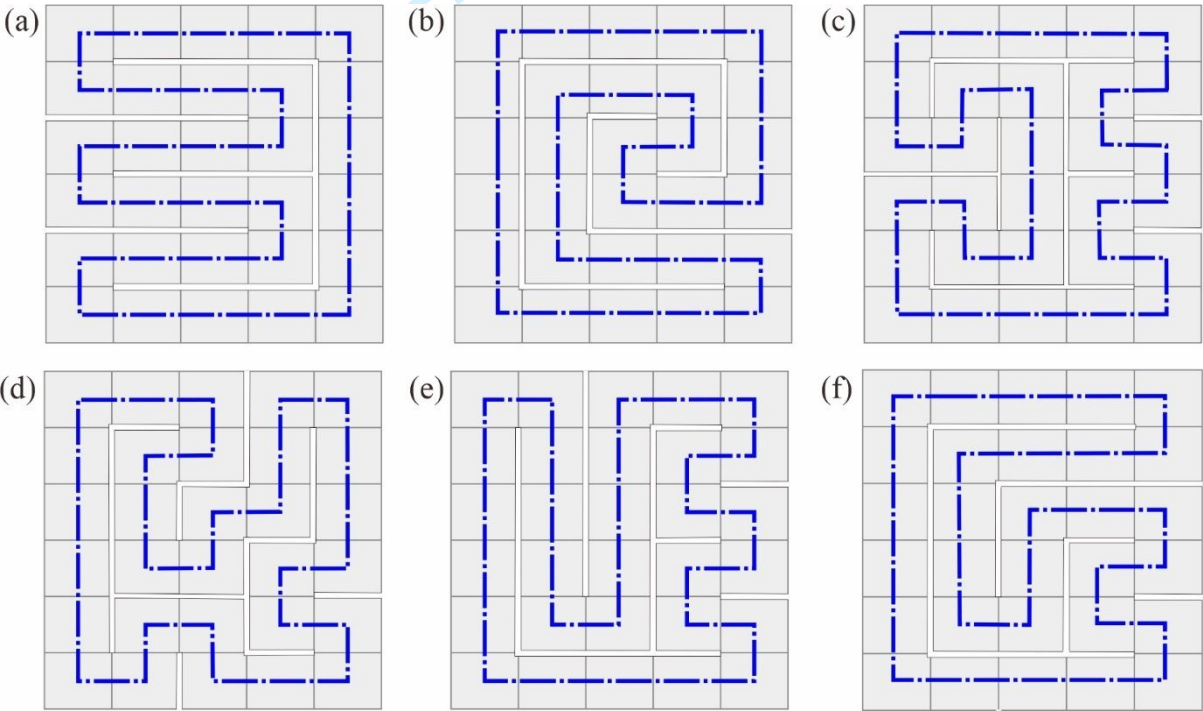


Fig. 5 Six HCs are enumerated in 6×5 rectangular arrays. Slits are highlighted accordingly.

2.2 Arrays with internal holes

The HC approach applies not only to regular tessellations but also to arrays with internal holes, provided the array consists of one type of polygon and each tile exhibits reflection symmetry along shared edges. Examples include a 3×3 array of squares with the central panel removed, and the mirror array of the James Webb Space Telescope. Additional examples are provided in the Supplementary Material.

3 Condition for folding arrays into two stacks

The existence of a HC does not necessarily guarantee a successful folding into two compact stacks. For example, the kirigami pattern in Fig. 5(f) always leads to clustering of certain panels, making it impossible to form two stacks with an equal number of panels. This leads us to the next question: *what is the condition for folding the array of panels into two equal stacks?*

Definition 3.1 (*Topology and geometry*) Consider an array of identical m -sided polygons whose sides are labelled as $\mathcal{S} = \{s_1, \dots, s_m\}$. They are connected sequentially by alternating mountain and valley folds to form a closed loop. Let the ordered set of polygons in the loop be denoted by $\mathcal{P} = \{P_1, \dots, P_i, \dots, P_{2n}\}$ and e_i be the shared edge between polygons P_{i-1} and P_i . Define

$$\mathcal{P}_{odd} = \{P_1, P_3, \dots, P_{2n-1}\} \text{ and } \mathcal{P}_{even} = \{P_2, P_4, \dots, P_{2n}\}. \quad (1)$$

Definition 3.2 (*Rigid Folding*) Folding along predefined creases (edges) without bending, stretching, or distorting the facets [32].

Consider the loop of $2n$ identical polygons. The loop contains $2n$ creases. Select one crease and make an imaginary cut along it. This transforms the loop into an open chain of polygons with two free ends, each connected to the selected imaginary cut-crease, now represented as two matching boundary edges. Starting from one end of the chain, fold the polygons sequentially in a zig-zag manner with alternating mountain and valley folds through $(n-1)$ creases. This results in one stack of n polygons. Apply the same zig-zag folding procedure from the other end of the chain. This produces a second stack, also with n polygons. If the two cut-edges coincide after this process, i.e., they overlap with matching orientation, then the stacks can be achieved without physically cutting the loop open, and the folding can be completed by reconnecting the cut-edges without conflict. In this case, the loop folds into two equal stacks, with each stack containing half numbers of the polygons. If not, the selected crease must not be on the outer layer of the stacks and the loop cannot be folded into two equal stacks.

Lemma 3.1 A loop of identical polygons is possible to be rigidly folded into two equal stacks if there exists a shared edge along which the loop can be virtually cut to form an open chain, such that folding from both ends of the chain in a zig-zag sequence leads to the two cut-edges coinciding after an equal number of folds.

Proposition 3.1 (*Vector reflection condition*) The zig-zag foldings of panels can be regarded as reflection operations about their common creases on a two-dimensional surface. In the closed loop of polygons, making an imaginary cut along edge e_i transforms the Hamiltonian circuit into a Hamiltonian path, with the original edge e_i now split into two labelled vectors, e_i^- and e_i^+ . Folding proceeds from each end of the Hamiltonian path in an alternating sequence of mountain and valley folds, corresponding to successive reflections. If, after $n-1$ reflections from both ends, the transformed edge vectors e_i^- and e_i^+ can be identified with the same orientation, the loop of polygons can be folded into two equal stacks.

Proof 3.1 If the same number of reflections from both ends map e_i^- and e_i^+ onto each other with the same position and orientation, the polygons at the open ends of the Hamiltonian path can be folded back to rejoin and close the loop. This implies that the loop of polygons can fold into two stacks with equal number of panels. $\square$

Example 3.1 Let us apply the vector reflection method to a 6×5 array, in which slits have been made according to the HC plotted in Fig. 5(e). There are 30 panels in this array and are labelled (Fig. 6a). If an imaginary cut is made along the crease that is connecting P_1 and P_2 , the closed loop becomes an open chain, with the red edges being on both ends of the chain (Fig. 6b). Next, panel P_1 is folded about creases in a zig-zag manner sequentially, and so is panel P_2 . All

panels are now folded onto two panels, P_{16} and P_{17} . It turns out that the imaginary cut-edges do not coincide in the end (Fig. 6c), suggesting that the red crease must be physically cut open to achieve the two-stack configuration.

When we cut open the loop along the red crease that connects panels P_2 and P_3 (Fig. 6d), P_2 and P_3 are folded onto P_{18} and P_{17} , respectively, after a series of zig-zag folding (Fig. 6e). Now the red edges coincide in the folded stacks, demonstrating that the array can be folded into two stacks in parity and the crease that connects panels P_2 and P_3 lies on the outer layer of the folded stack. The configuration in Fig. 6(e) is physically realisable without requiring a cut.

Mathematically, the zig-zag foldings of panels are regarded as reflection operations about their common creases. We name the creases as e_i ($i = 1, 2, \dots, 30$) as shown in Fig. 7(a). When crease e_2 is chosen to be the cut crease, it is denoted by two identical vectors, $\mathbf{e}_2^+$ and $\mathbf{e}_2^-$ at the two ends of the open chain, respectively (Fig. 7b). The vector $\mathbf{e}_2^+$ is reflected about the edges $e_3, e_4, \dots, e_{16}$ in sequential order, with each operation being tracked by black arrows. Likewise, the vector $\mathbf{e}_2^-$ is reflected about the creases $e_1, e_{30}, \dots, e_{18}$ in succession. After 14 operations on each open chain, cannot be identified at the same location and orientation, indicating that the crease e_2 must not be on the top layer of the folded stack. When e_3 is selected to be the crease to cut open instead, the vectors $\mathbf{e}_3^+$ and $\mathbf{e}_3^-$ are reflected sequentially (Fig. 7c). They coincide at crease e_{18} after 14 operations for each of the two stacks, indicating that the array is possible to be folded in two stacks with the top and bottom layers being P_{18}, P_{17} and P_2, P_3 , respectively. The condition stated in Proposition 3.1 is satisfied.

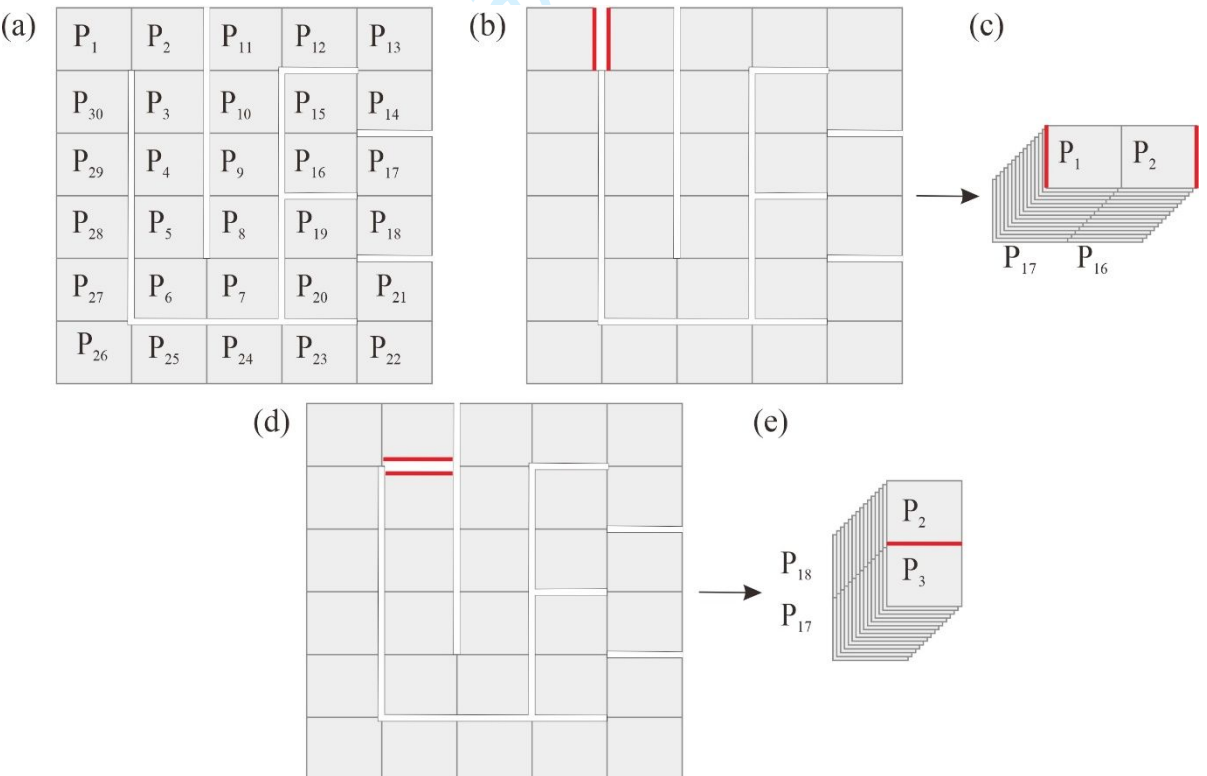


Fig. 6 Folding a 6×5 rectangular array into two stacks. (a) Panels are labelled. (b) - (c) A crease that does not satisfy, and (d) - (e) a crease that satisfies the two-stack foldable criterion.

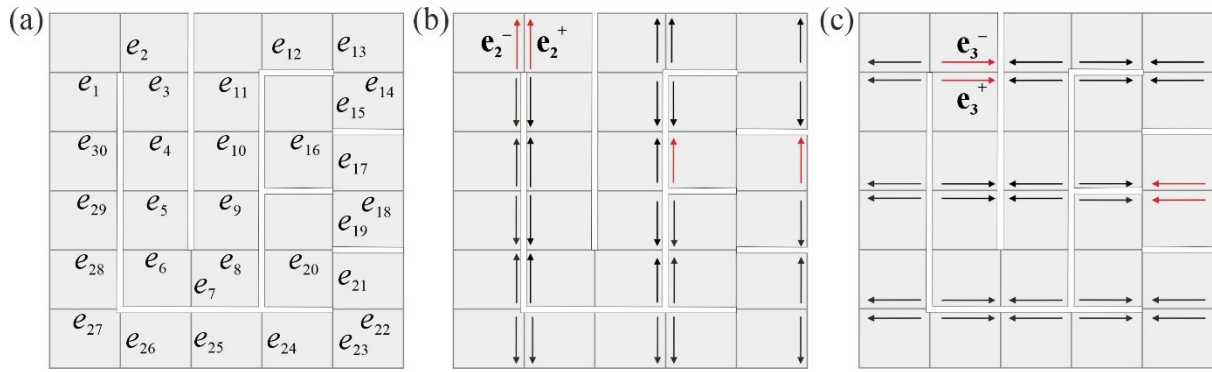


Fig. 7 Zigzag foldings are regarded as reflection operations. (a) Creases are labelled in a 6×5 rectangular array. (b) Vectors e_2^+ and e_2^- do not coincide, and (c) vectors e_3^+ and e_3^- coincide, each after 14 reflection operations.

Remark 3.1 By cutting the loop of polygons open, the problem of folding a two-dimensional sheet is reduced to the zig-zag folding of a one-dimensional chain. Although the crease axes in the chain may not be parallel, determining whether an open chain can be folded into two equal stacks is significantly simpler than analysing the original closed-loop spatial linkage.

However, satisfying Proposition 3.1 alone is not enough to judge if a closed-loop kinematic chain can be folded into two stacks with an equal number of panels because we cut the loop of polygons open and rejoined it back.

Example 3.2 A configuration that satisfies Proposition 3.1 but fails to fold into two stacks

In an array of 12 regular hexagons (Fig. 4i), slits are introduced according to a HC. The resulting kirigami pattern is duplicated, with labels beginning at the polygon in the top-right corner of the array and proceeding sequentially along the loop (Fig. 8a). The creases are labelled similarly. Applying the vector reflection method (Proposition 3.1), we trace five consecutive reflections (indicated by black arrows in Fig. 8b) to identify the vectors e_6^+ and e_6^- . This indicates that the polygons P_5 and P_6 are possible candidates to fold onto P_{12} and P_{11} , respectively. The folding behaviour was tested using a physical card model with two distinct face colours – shaded grey on the top facet and light yellow on the bottom (Fig. 8c). The array failed to fold into two compact stacks as intended. Instead, it formed an entangled loop with self-interference during the folding process, preventing successful stacking (Fig. 8d). Interestingly, if the loop is cut open along e_6 with the two open ends twisted by 2π and then the ends are glued back together, the loop of polygons can be folded into two stacks with an equal number of panels.

In this example, the prototype was made by cards instead of thick panels to demonstrate the entanglement in the folding process. If an entanglement in the card model is inevitable, it is not resolvable in the thick-panel model as more constraints are imposed in a loop of rigid panels.

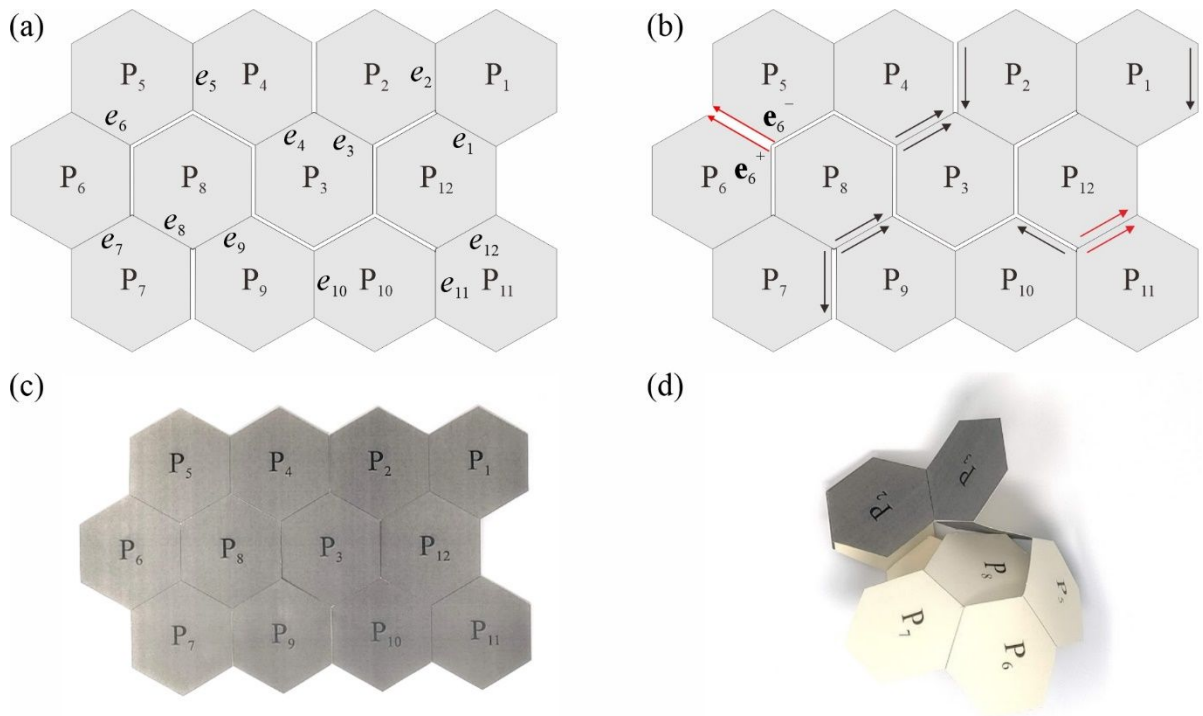


Fig. 8 In an array of 12 regular hexagons, (a) Hexagons and creases are labelled in the loop. (b) e_6^+ and e_6^- can be identified with the same location and orientation after vector reflections. (c) A card model at deployed state. (d) The panels fail to fold into two stacks.

4 Condition for folding a closed loop of polygons without entanglement

We first consider a loop of polygons as a continuous band. By taking a bi-coloured open loop and identifying its two ends with vectors of the same direction, we can obtain a closed loop topologically equivalent to a cylinder (Fig. 9a). In this configuration, no relative rotation occurs between the two ends before closure. If instead the two ends of the same bi-coloured open chain are identified with vectors of opposite direction, the resulting surface is a Möbius strip (Fig. 9b). This configuration introduces a turn of π about the neutral axis of the chain before closure (also known as a half twist) and results in a single-sided surface. Topologically, a cylinder and a Möbius strip are distinct, but a turn of 2π about the neutral axis of a chain (also known as a full twist) is homeomorphic to a band with no twists [33]. Such homeomorphism can be demonstrated in higher dimensions by cutting the band, untwisting it, and rejoining the edges, as long as the two edges are identified in the same direction – similar to what we have done in Section 3. However, such homeomorphism cannot be extended to 3-dimensional Euclidean space ($\mathbb{R}^3$), where a cut cannot be made physically. Therefore, we need to find a method to characterise twists in $\mathbb{R}^3$.

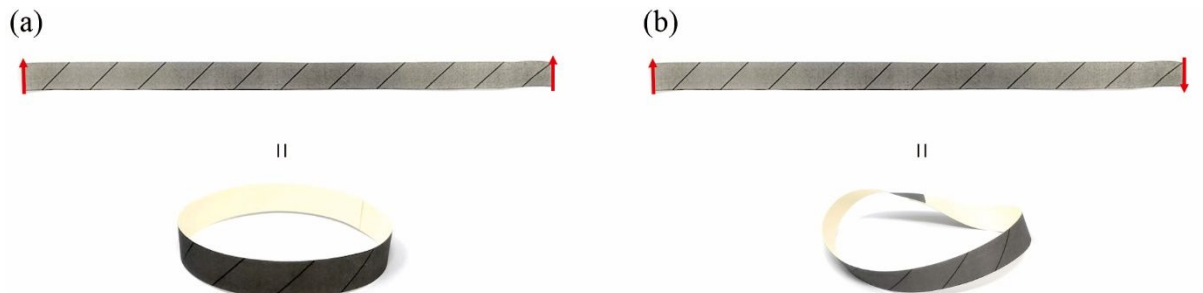


Fig. 9 (a) A cylinder, (b) a Möbius strip.

4.1 Characterising the entanglement – twist and writhe

We borrow the concepts from topology of space curves and knot theory to describe the geometry of the loop of polygons using twist (Tw) and writhe (Wr). Twist measures the cumulative local rotation of the facets around the neutral axis of the loop of panels. It encodes the contribution of local folding to the spatial configuration in $\mathbb{R}^3$. Writhe measures the coiling of the loop's neutral axis itself, treating the loop as a one-dimensional spatial curve. According to the Călugăreanu-White-Fuller theorem [34], the twist resulted from accumulated local rotations can be transferred to the global writhe, while the sum of Tw and Wr is constant:

$$Lk = Wr + Tw. \quad (2)$$

where Linking number (Lk) is a topological invariant representing the total number of times one edge of the folded strip winds around the other.

Example 4.1 A once-twisted band can be formed by turning one end of a bi-coloured band by 2π about its neutral axis before joining it back to the other end (Fig. 10a, b). In the configuration immediately before joining the two ends, the deformation is stored entirely as twist, characterised by $Tw = 1$ and $Wr = 0$ (Fig. 10c). By physically manipulating the band, the local twist can be redistributed into spatial coiling of the band's neutral axis, where now $Tw = 0$ and $Wr = 1$ (Fig. 10d). Throughout the manipulation, the total linking number remains unchanged.

Lemma 4.1 If nonzero twist is introduced during the folding process, the loop will be entangled and cannot be folded into two stacks.

Proof 4.1 For a loop of polygons initially in a flat configuration, the linking number is $Lk = 0$ and remains unchanged throughout the folding process. The folding motion may induce local rotations between adjacent facets, therefore introducing nonzero Tw. Meanwhile, as no global coiling of the loop's neutral axis itself is introduced during the folding motion, Wr remains zero. According to the Călugăreanu-White-Fuller theorem, the relation $Lk = Wr + Tw$ must be satisfied. Hence, any accumulation of twist would imply a nonzero linking number in the folded state, contradicting the initial condition $Lk = 0$. This implies that the loop of polygons becomes entangled during folding, making the kirigami pattern non-foldable.

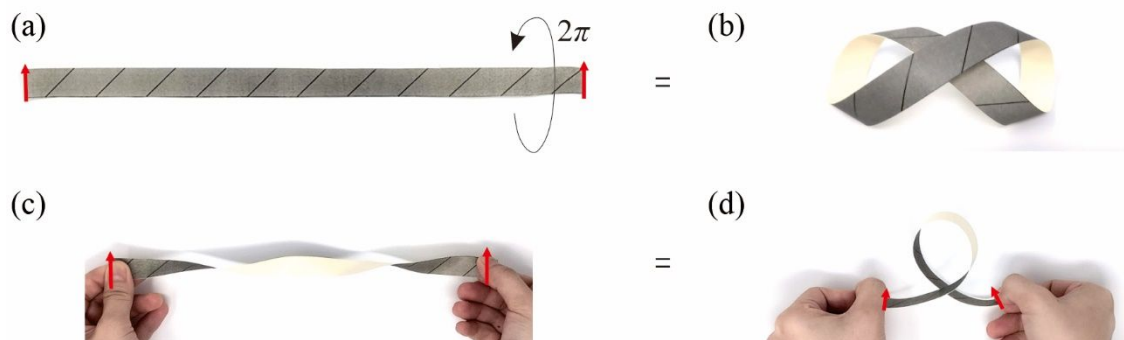


Fig. 10 A band with a full twist: (a) One end of a bi-coloured band is turned by 2π and identify with the other end with the same orientation. (b) A closed band with a full twist. (c) Before closure: $Tw = 1$, $Wr = 0$. (d) After physical manipulation: $Tw = 0$, $Wr = 1$.

Remark 4.1 The problem of determining whether a closed loop of polygons will encounter an entanglement during the folding process is reduced to assessing if a non-zero twist exists. To characterise the twist, we calculate the cumulative local rotations of the panels induced by folding motions.

4.2 Local rotations induced by folding

Folding a polygon onto its connected neighbour is geometrically equivalent to reflecting it across the shared crease. When a polygon is successively reflected across two adjacent creases, the combined transformation is a rotation about the intersection point of the creases. The rotation angle equals twice the angle between the two creases. This outcome follows from Euclidean isometry: a composition of two reflections across intersecting lines results in a rotation, while reflections across parallel lines produce a translation [35]. In the context of folding loops of panels, two consecutive reflections introduce local rotations that accumulate as twist around the loop's neutral axis.

Example 4.2 Consider a loop of regular hexagons where P_i is connected to P_{i+1} via side s_6 , and P_{i+1} is connected to P_{i+2} via side s_2 (Fig. 11). Let α denotes the angle between sides s_6 and s_2 . The polygon P_i is first reflected across side s_6 to coincide with P_{i+1} and then reflected again across one side of P_{i+1} to coincide with P_{i+2} . There are five possible choices for the second reflection (from s_1 to s_5), leading to five configurations for any three connected hexagons in the loop.

Definition 4.1 (*Rotation resulted from reflections*) Define a rotation resulted from reflecting an m -sided polygon P_i onto P_{i+1} , followed by a second reflection onto P_{i+2} as

$$\gamma_i = 2\alpha_i, \gamma_i \in [-2\pi, 2\pi], \quad (3)$$

where $i = \{1, 2, \dots, 2n\}$ noting that $2n + 1$ becomes 1 and $2n + 2$ becomes 2 for a closed kinematic chain. The first reflection is across the shared edge s_m of P_{i+1} , and the second reflection is across one of the edges $\{s_1, s_2, \dots, s_{m-1}\}$ of P_{i+1} .

This can be further illustrated by the connected hexagons in Example 4.2. When the first reflection is across side s_6 , the resulting rotation angle γ_i depends on the second reflection across P_{i+1} : $4\pi/3$ for s_1 ; $2\pi/3$ for s_2 ; 0 for s_3 ; $-2\pi/3$ for s_4 ; $-4\pi/3$ for s_5 .

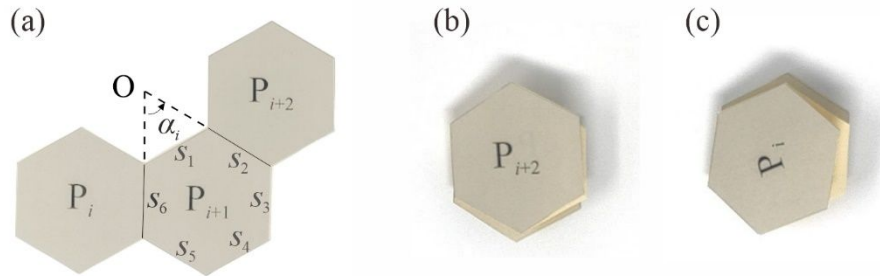


Fig. 11 Folding a physical card model of three consecutive regular hexagons. (a) The first reflection is across s_6 , followed by s_2 of P_{i+1} . (b) A valley followed by a mountain fold results in P_i underneath P_{i+2} . (c) Reversing the folding sequence results in P_i on top of P_{i+2} .

Note that the listed rotation angle γ_i resulted from two consecutive reflections does not capture the folding directions, i.e., it does not distinguish between mountain or valley folds.

Consider the case in Example 4.2. When P_i is folded onto P_{i+1} via a valley fold along s_6 , followed by folding P_{i+1} onto P_{i+2} via a mountain fold along s_2 , P_i is rotated γ_i and ends up underneath P_{i+2} (Fig. 11b). If the folding sequence is reversed, i.e., a mountain followed by a valley fold, P_i is still rotated by γ_i but lies on top of P_{i+2} (Fig. 11c). Therefore, if the strip were closed into a loop, these two configurations would result in opposite twist directions of the loop.

To account for the fold direction, we have

Definition 4.2 (Local rotation) We assume that all creases are valley folds when $i = 1, 3, \dots, 2n - 1$, whereas all creases are mountain folds when $i = 2, 4, \dots, 2n$. The local rotation resulting from folding P_i onto P_{i+1} and then folding onto P_{i+2} is defined as

$$g(i) = \sigma_i \cdot \gamma_i, \quad (4)$$

where the sign factor σ_i is given by

$$\sigma_i = \begin{cases} +1 & \text{if } i = 1, 3, \dots, 2n - 1; \\ -1 & \text{if } i = 2, 4, \dots, 2n. \end{cases} \quad (5)$$

Definition 4.3 (Cumulative Rotations) The cumulative local rotations in the loop are

$$Odd(\mathcal{P}) = \sum_{i=1}^{i=2n} g(i) \text{ where } i = 1, 3, \dots, 2n - 1; \quad (6)$$

$$Even(\mathcal{P}) = \sum_{i=1}^{i=2n} g(i) \text{ where } i = 2, 4, \dots, 2n.$$

$Odd(\mathcal{P})$ and $Even(\mathcal{P})$ represent the sum of local in-plane rotations associated with odd- and even-indexed polygons, respectively.

Likewise, the local rotation induced by folding a loop of squares or equilateral triangles is illustrated in Fig. 12. For squares, each polygon has four sides. When P_i is first folded onto P_{i+1} across s_4 of the square, the resulting rotation angle γ_i depends on the second reflection across P_{i+1} : π for s_1 , 0 for s_2 , and $-\pi$ for s_3 (Fig. 12a). For equilateral triangles with three sides, folding P_i onto P_{i+1} across s_3 followed by across s_1 yields $\gamma_i = 2\pi/3$, and $\gamma_i = -2\pi/3$ when the second reflection is across s_2 (Fig. 12b). The sign factor of the rotation can be determined based on whether i is even or odd.

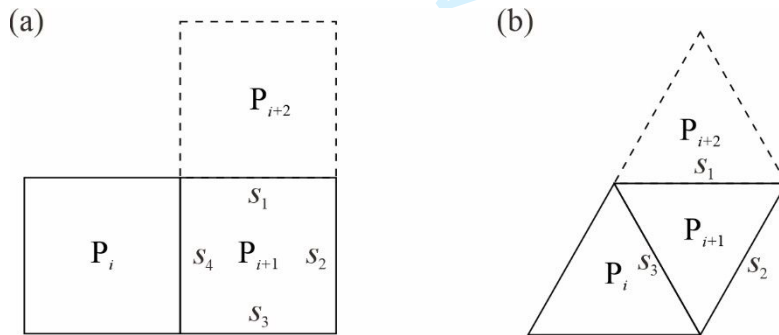


Fig. 12 Folding three consecutive (a) squares and (b) equilateral triangles.

The cumulative effect of local in-plane rotations determines the twist accumulated during the rigid folding of a loop of polygons. For instance, folding P_1 onto P_2 and then onto P_3 induces a local rotation $g(1)$. Similarly, folding P_2 onto P_3 and then onto P_4 yields another local rotation $g(2)$. However, note that the folding from P_2 onto P_3 appears in both steps and is therefore double counted. Continuing the process around the loop, each fold between adjacent polygons P_i and P_{i+1} is included in two local rotations.

To correct this overcounting, we sum the local rotations from all steps, i.e., $Odd(\mathcal{P}) + Even(\mathcal{P})$, and then divide by two. Finally, normalising the result by 2π yields the expression for the net twist, which captures the total rotational effect in units of full twist.

Definition 4.4 (Twist) The accumulated twist by folding a closed loop of polygons is

$$Tw = \frac{1}{4\pi} \sum_{i=1}^{i=2n} g(i). \quad (7)$$

where $g(i)$ is the local in-plane rotation angle when folding P_i onto P_{i+1} and then onto P_{i+2} .

Proposition 4.1 A loop of polygons can be rigidly folded onto two stacks without entanglement during the folding process if and only if

$$Tw = \frac{1}{4\pi} \sum_{i=1}^{i=2n} g(i) = 0. \quad (8)$$

Proof 4.2 Necessary condition Following Lemma 4.1, no global entanglement requires zero twist introduced in the folding process, i.e., $Tw = 0$. From Definition 4.4, the net twist is formulated as in Eqn. (7). **Sufficient condition** Conversely, we assume $Tw = 0$. Since Tw is the accumulated in-plane rotation from local folds, a zero value implies that the rotational angles are balanced for even-indexed polygons and odd-indexed polygons across the loop, indicating that the folded shape is neither twisted nor entangled. $\square$

This result provides a systematic method to verify foldability: by summing the local rotations across all folding steps, we can predict whether a given crease pattern will lead to a two-stack configuration free from physical entanglement.

Example 4.3 Now we apply Proposition 4.1 to a few kirigami patterns and calculate their accumulated rotations for all even-indexed and odd-indexed polygons to assess their foldability. For the entangled loop of hexagons in Example 3.2, we have $Odd(\mathcal{P}) = 0$ and $Even(\mathcal{P}) = -4\pi$, indicating that a once-twisted loop is introduced in the folding process. This agrees with our earlier observations (Fig. 8). Some more kirigami patterns are shown in Fig. 13. In the array of squares with an internal hole (Fig. 13a), $Odd(\mathcal{P}) = 0$ and $Even(\mathcal{P}) = 4\pi$, resulting in a twisted loop in the folding process. In the array of squares in two rows and four columns (Fig. 13b), $Odd(\mathcal{P}) = -2\pi$ and $Even(\mathcal{P}) = 2\pi$, indicating that the two-stack folding is possible. So does the array of equilateral triangles (Fig. 13c). It is possible to be folded into two stacks following Proposition 4.1, yielding $Odd(\mathcal{P}) = -2\pi$ whereas $Even(\mathcal{P}) = 2\pi$.

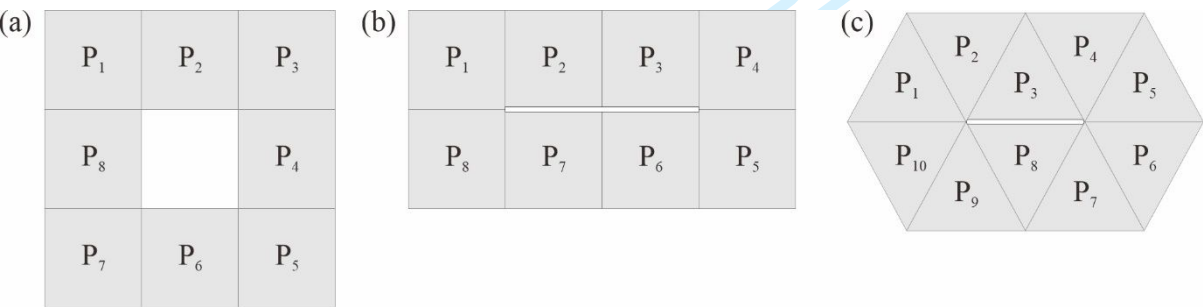


Fig. 13 Kirigami patterns: (a) Squares with an internal hole; (b) tessellation of squares; (c) tessellation of equilateral triangles.

5. Necessary and sufficient conditions to fold thick panels into two stacks

Theorem. A loop of identical polygonal panels with uniform thickness can be rigidly folded into two stacks if and only if: (i) these polygons can be folded into two equal stacks under

vector reflection condition (Proposition 3.1), and (ii) the folding process introduces no twists (Proposition 4.1).

Proof. Necessary condition Suppose a loop of panels can be rigidly folded into two compact stacks with equal number of panels. Then the shared edge on the top layer of the stacks aligns with the shared edge on the bottom layer of the stacks. In other words, the vector along the shared edge shall be reflected counterclockwise and clockwise for an equal number of times and be identified at the same location and orientation, as described in Proposition 3.1. Moreover, the folding must be feasible without entanglement and twist, implying $Tw = 0$ and therefore Proposition 4.1 holds. Both conditions are necessary. **Sufficient condition** Conversely, assume that the loop satisfies both Proposition 3.1 and 4.1. The former ensures that after cutting the loop and folding from both ends of the resulting Hamiltonian path, the two open edges of the loop align after the same number of folds. This alignment allows the loop to be joined back to a closed loop and ensures that the polygons are evenly divided between the two resulting stacks. The latter ensures that without cutting the loop open, the folding process is free from entanglement and therefore feasible to be rigidly folded into two stacks. Together, these two conditions imply that the loop can be rigidly folded into two equal stacks. $\square$

For kirigami patterns that satisfy the conditions outlined in the Theorem, the polygons on the top and the bottom layers of the stacks can be found. They correspond to the two polygons sharing the cut-edge e_i that satisfies the parity condition. A programme has been developed based on the mathematical framework established in the Theorem to support practical engineering applications. Users can specify a deployed shape composed of identical tiles (polygonal panels) arranged in x rows and y columns. The programme then generates a kirigami pattern that enables the user-defined shape to fold into two stacks, each containing an equal number of tiles. Further details about the programme are provided in the Supplementary Material.

Example 5 A configuration that satisfies both conditions In an array of ten regular hexagonal panels, two HCs can be found – one generating an asymmetrical kirigami pattern (Fig. 14a) and a symmetric one (Fig. 14b). By placing slits and creases along the HCs, the flat surface becomes a loop of ten connected hexagons. In the first kirigami pattern (Fig. 14a), the regular hexagons can be labelled as P_i ($i = 1, 2, \dots, 10$), with P_1 at the top right corner (Fig. 14c). The creases that connect the hexagons are denoted e_i .

First, we check again Proposition 3.1, the vector reflection condition. We start searching for the crease that is on the outer layer of the folded stacks from the edge e_1 . The cut-edges that are attached to the two ends of the open chain are denoted as two identical vectors, $\mathbf{e}_1^+$ and $\mathbf{e}_1^-$. Vector $\mathbf{e}_1^+$ is reflected about the edges e_2, e_3, e_4 and e_5 in sequential order. Likewise for the vector $\mathbf{e}_1^-$, the reflection operations are about the creases e_{10}, e_9, e_8 and e_7 in succession. Each after four operations, the red vectors cannot be identified at the same location, indicating that the crease e_1 is not on the outer layer of the folded stack. The searching continues until edge e_4 . Two vectors $\mathbf{e}_4^+$ and $\mathbf{e}_4^-$ are attached on P_4 and P_3 , respectively, marked in red in Fig. 14(d). After a series of reflection operations that are tracked by black arrows, the two red vectors coincide at crease e_9 , indicating that the array is possible to be folded in two stacks with the top and bottom layers being polygons P_4, P_3 and P_8, P_9 . Likewise, when edge e_5 is investigated and the two vectors $\mathbf{e}_5^+$ and $\mathbf{e}_5^-$ are attached on P_5 and P_4 , respectively (Fig. 14f), the two red vectors coincide at crease e_{10} after a series of reflection operations.

Next, we evaluate the kirigami pattern against Proposition 4.1, the zero-twist condition. The accumulated rotation of the odd-indexed hexagons is 2π , and that of the even-indexed hexagons is -2π , indicating that the folding process introduces no net twist in the loop of hexagons.

Satisfying both conditions, the array of hexagonal panels that is connected based on the HC in Fig. 14(a) can be folded into two compact stacks. A thick-panel prototype was made to demonstrate the two-stack foldability of the array. In Fig. 14(e), panels P_3 and P_4 fold onto P_9 and P_8 , with all the other panels sandwiched in-between. In Fig. 14(g), panels P_{10} and P_9 were folded onto P_4 and P_5 , respectively.

The kirigami pattern generated from the symmetric HC in Fig. 14(b) does not satisfy the zero-twist condition and therefore, cannot be folded into two stacks.

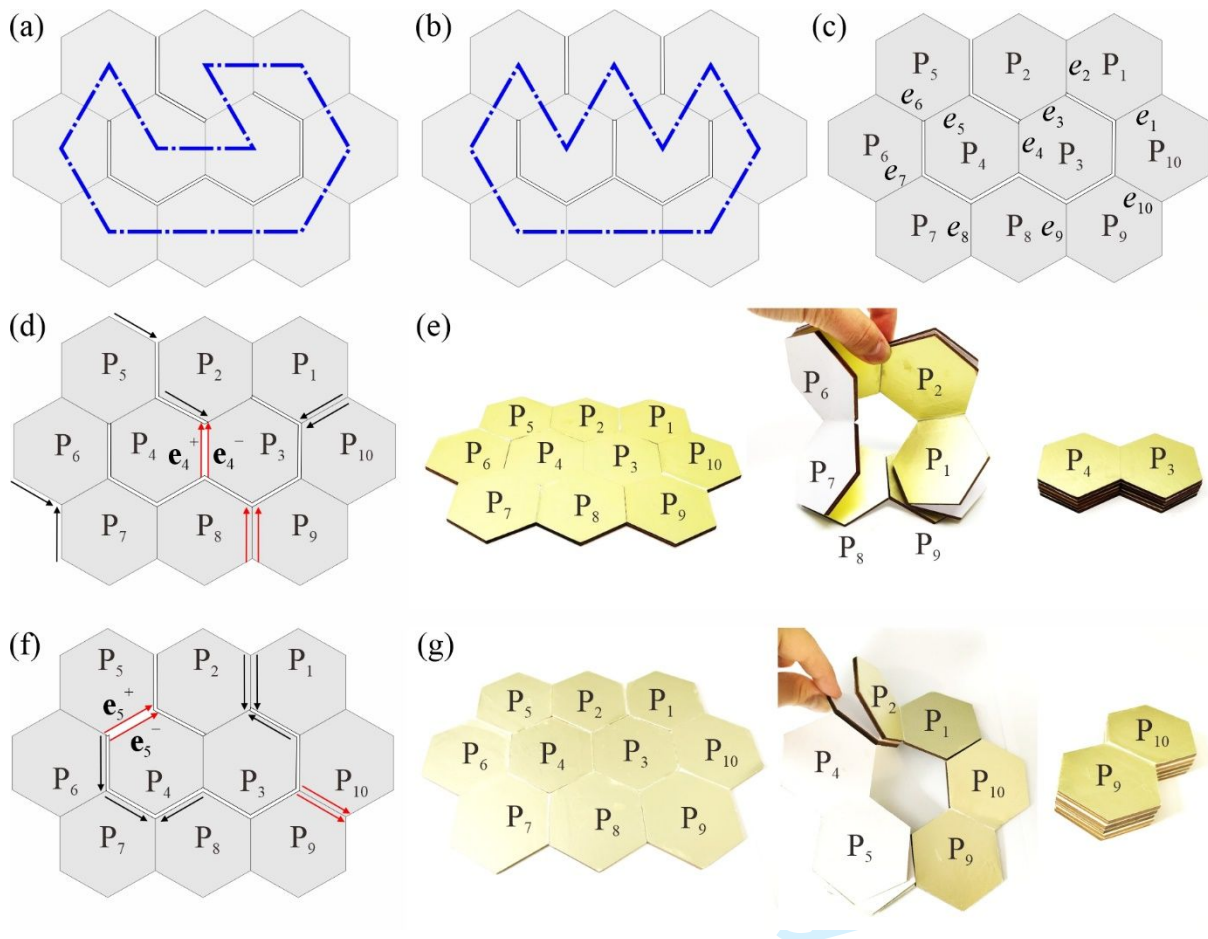


Fig. 14 In an array of ten regular hexagons, (a) an asymmetric HC and (b) a symmetric HC are drawn with dotted dash lines. (c) Panels and edges are labelled in the array according to the asymmetric HC. (d) The vectors e_4^+ , e_4^- satisfy the two-stacks foldable criterion. (e) Folding of a thick-panel prototype to confirm the two-stack foldability. (f) The vectors e_5^+ , e_5^- also satisfy the two-stacks foldable criterion for the same HC. (g) Folding of a physical prototype.

6. Rigidising creases

For the loops of polygonal panels that can be folded into two stacks in parity, there is no relative motion between panel pairs in the top and bottom layers. Consequently, the creases on these outer layers can be rigidised without affecting the loop's foldability and stackability. Moreover, for the feasible folding patterns that exhibit symmetry, certain slits along the plane of symmetry may be unnecessary. In such cases, adjacent panels can be merged along the symmetry axis to form larger panels, doubling their size relative to the originals.

Example 6.1 Among the kirigami patterns in a 6×5 array of squares, the symmetrical one shown in Fig. 5(a) satisfies both conditions and therefore can be folded into two compact stacks. A physical model was made based on the symmetric HC and reproduced in Fig. 15(a). In the fully folded configuration, the pair of panels at the centre-left appears at the top of the stacks whereas the pair of panels at the centre-right is at the bottom. Both pairs of panels are illustrated in darker grey. The creases between these two pairs of panels need not be activated as explained previously. Moreover, due to the symmetrical geometry, the slits along the plane of symmetry are also unnecessary. The five pairs of panels on the two rows of the array can be merged to form five larger panels twice the size of the original rectangular panels. The folding sequence of the physical model is given in Fig. 15(b). No voids are observed in the fully packaged state.

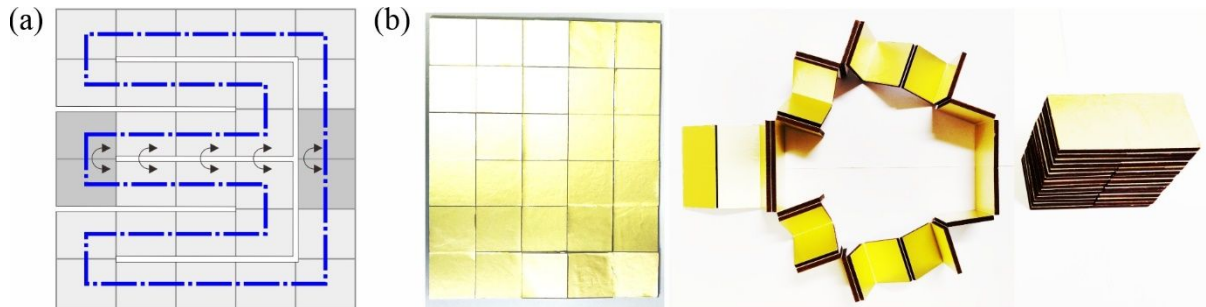


Fig. 15 A 6×5 rectangular array with symmetry. (a) Panels are merged along the plane of symmetry. (b) Folding process of a thick-panel prototype.

Using the Kutzbach criterion for a spatial closed-loop linkage with revolute joints, the DoF is $6(N-1) - 5J$ where N is the number of links and J is the number of revolute joints. In a 6×5 array, $N = 30$ and $J = 30$ after slits have been made, resulting in a 24-DoF assembly. The large number of DoF is not preferred since many actuators will be required to control the deployment process. For symmetric HCs, some panels along the plane of symmetry could be merged as shown in Fig. 5(a). The DoF of the 6×5 array is reduced from 24 to 19 for the design.

Despite merging some panels which reduces the DoF of an assembly, the number of DoFs remains high. Deploying such a multi-DoF structure can be achieved either through active control using multiple actuators or via passive deployment strategies. Examples of the latter include tape springs, linear torsional springs, and shape memory materials, which enable synchronised motion without heavy and cumbersome control systems.

7. Conclusion

This paper developed a mathematical framework that integrates geometry, graph theory, and rigid-foldable origami and kirigami principles to enable the folding of a tessellated planar surface with uniform thickness into two compact stacks. By identifying the HCs within the tessellated grid, we derive the kirigami patterns and establish necessary and sufficient conditions for rigidly folding the resulting thick-panel kirigami model into two compact stacks. Based on the mathematical framework, an open-source programme has been developed for practical engineering applications. These results were validated through analytical proofs and physical prototypes. In addition, we proposed design strategies to reduce the system's DoFs, thereby simplifying the deployment process and improving the mechanical rigidity in the deployed state. This work advances the theoretical understanding of kirigami-inspired deployable systems and provides practical guidelines for designing compact, rigid foldable structures made from identical interlinked panels with uniform thickness.

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While this work focuses on folding planar structures into two-stack configurations, the proposed framework can be extended to design any even number of stacks by combining multiple two-stack modules. With minor geometric modifications to the uniform-thick panels, the method can also be adapted to deploy a compact volume to approximate a curved surface. Further research directions include folding into asymmetric or odd-numbered stacks. Additional opportunities include extending the framework to handle non-identical polygonal tessellations and actuating multi-DoF deployable systems using active materials such as shape memory materials.

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